EMNLP BoF 2026 · October 27 · 9:00–10:30 AM

# Let's talk about:
# Symbolic and Trustworthy Reasoning Across Language and Vision

**Panel · Lightning talks · Open discussion**

Organized by Shreya Rajpal, Michigan State University

EMNLP2026 BUDAPEST

EMNLP BoF 2026 · October 27 · 9:00–10:30 AM

# How and when can structured and symbolic methods make reasoning more verifiable and trustworthy?

**Panel · Lightning talks · Open discussion**

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EMNLP2026 BUDAPEST

# Symbolic reasoning and trustworthiness

## Symbolic reasoning

Knowledge is encoded as explicit symbolic structures (entities, relations, rules). Reasoning applies rules to these structures to derive new ones.

`left(A,B) ∧ left(B,C)`
`⇒ left(A,C)`

rule: transitivity

## Trustworthiness

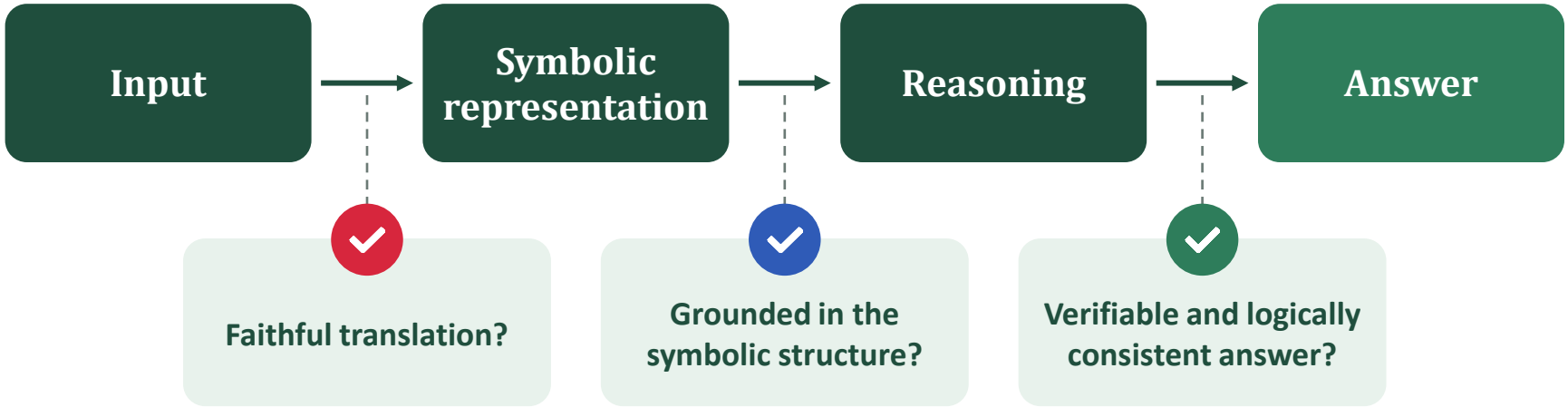
A model's answer is trustworthy when it's reasoning is based on the right evidence and is logically coherent.

**Faithful:** reflects how the model actually reached the answer

**Plausible:** is coherent and convincing to a human reader

*Sources: Newell & Simon (1976), Computer Science as Empirical Inquiry: Symbols and Search, CACM · Jacovi & Goldberg (2020), Towards Faithfully Interpretable NLP Systems, ACL · Shailya, Rajpal, Krishnan & Ravindran (2025), LExT: Towards Evaluating Trustworthiness of Natural Language Explanations, FAccT*

# Symbolic structure for trustworthy reasoning




**Explicit symbols make each step inspectable,** so faithfulness can be checked at every step, not only at the final answer.

# Quick show of hands



**You build structured or symbolic reasoning systems**



**You work on making model reasoning trustworthy**



**You use structure to make reasoning trustworthy**

# Send us your questions

## Scan to submit a question for the panel or the speakers

We'll pick up submitted questions during the open discussion.



# Today's program



**9:00**
Welcome & panel introductions

**9:06**
Opening lightning talk

**9:15**
Three lightning talks

**9:45**
Panel discussion

**10:15**
Audience Q&A & open discussion

*Ends 10:30 · Questions welcome throughout*

# Our panel



**Daniel Preotiuc-Pietro**

Bloomberg

**Structured reasoning & calibration**



**Leslie Barrett**

Bloomberg Law

**Faithful & structured reasoning**



**Parisa Kordjamshidi**

Michigan State University

**Language, vision & symbolic reasoning**



**Artem Shelmanov**

MBZUAI

**Uncertainty & reliable LLMs**

# Lightning talks

1

**LEGO: Synergizing Expert GraphRAG and Expert Chain-of-Thought for Legal Reasoning**

**Qingjing Chen**

PhD candidate, University of Bologna
EMNLP Findings 2026

2

**Transforming Interpretability Gains to Explainability Wins**

**Gokul S Krishnan**

Senior Research Scientist, Centre for Responsible AI, IIT Madras

3

**Grammar of the Wave: Towards Explainable Multivariate Time Series Event Detection via Neuro-Symbolic VLM Agents**

**Sky Chenwei Wan**

PhD Student, SLB, France
EMNLP 2026 (Main)

*5 min talk + 5 min discussion each*

Opening lightning talk

# When does structure make reasoning trustworthy?

Shreya Rajpal · Michigan State University

1 Reasoning in natural language

2 Symbolic translation

3 Results

4 Other applications

# Reasoning with natural language

**Story**

A is at 3 o'clock from N. B is above A. S is south of B. G is east of S.

**Where is G relative to N?**

→

LLM REASONS IN TEXT

*"…G falls between 12 and 9 o'clock, so G is slightly below N"*

→

✗ **G is below N**

hallucinated relation (answer: right of N)

## LESS ACCURATE AS COMPLEXITY GROWS

**93% → 27%**

accuracy from 1 to 10 hops (LLaMA-3.1-70B, StepGame)

## NOT TRUSTWORTHY: WHAT WE CHECK

| | |
|---|---|
| **Hallucinated relations** | **17%** of text failures |
| **Grounded in evidence (faithfulness)** | **52%** of answers |
| **Logically consistent (plausibility)** | **42%** of answers |

**Paper** *Rajpal, Premsri & Kordjamshidi. Spatial Reasoning via Modality Switching Between Language and Symbolic Representations. Findings of EMNLP 2026.*

# Translating text to symbolic representations

**Spatial story**

A is at 3 o'clock from N. B is above A. S is south of B. G is east of S.

**Question**

Where is G relative to N?

→

**Relational triples**

LLM extraction, sentence by sentence

| head | relation | tail |
|---|---|---|
| A | right | N |
| B | above | A |
| S | below | B |
| G | right | S |

→

**Full grid**

| | col 1 | col 2 | col 3 |
|---|---|---|---|
| row 1 | – | B | – |
| row 2 | **N** | A/S | **G** |

whole scene

**Coordinates**

```
N (0,0)   A (1,0)
B (1,1)   G (2,0)
```

Prolog constraint solver

**Pruned grid**

| | col 1 | col 2 |
|---|---|---|
| row 1 | – | – |
| row 2 | **N** | **G** |

question entities only

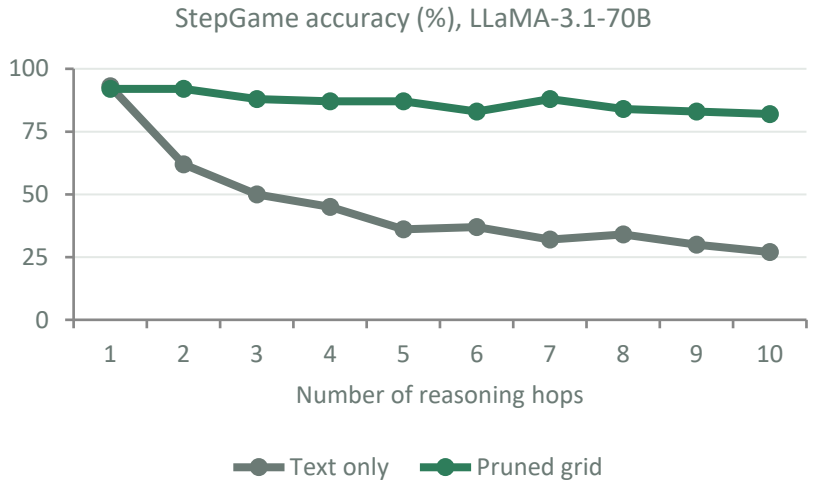
**Reasoning with triples**

*Grids and coordinates are built from the triples by code*

**Paper** *Rajpal, Premsri & Kordjamshidi. Spatial Reasoning via Modality Switching Between Language and Symbolic Representations. Findings of EMNLP 2026.*

# With symbolic representations

## STABLE ACROSS COMPLEXITY




## MORE ACCURATE, FEWER ERRORS

**+42%** accuracy

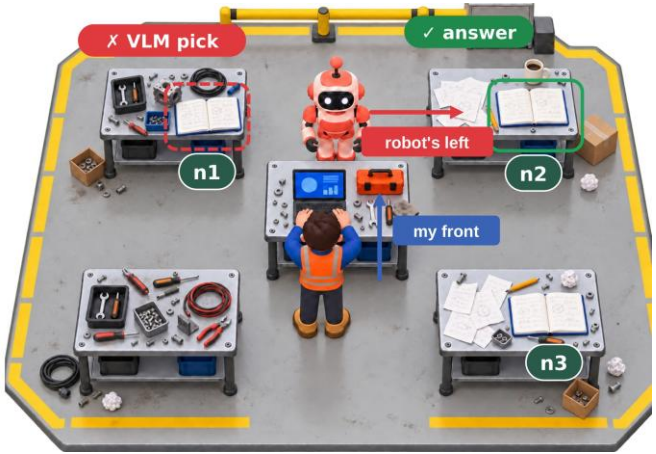
**64%** hallucinations fixed

**82%** Evidence-grounding errors fixed

**→ More trustworthy and more accurate reasoning**

**Paper** *Rajpal, Premsri & Kordjamshidi. Spatial Reasoning via Modality Switching Between Language and Symbolic Representations. Findings of EMNLP 2026.*

# Translating images to symbolic representations

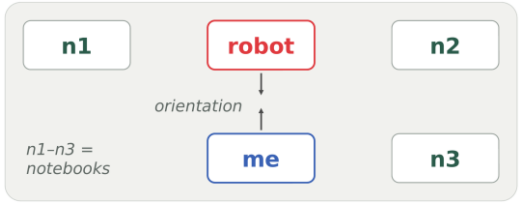



*"Bring me the notebook **on your left** and **in front of me**"*

**Reasoning in code**

**1** **3D scene parsing**

3D Positions + Orientations




**2** **Soft symbolic scores**

each relation in its own frame

| object | notebook | robot's left | my front |
|---|---|---|---|
| **n2** | **0.98** | **0.94** | **0.89** |
| n3 | 0.96 | 0.91 | 0.04 |

**3** **Symbolic program**

written by an LLM, then executed

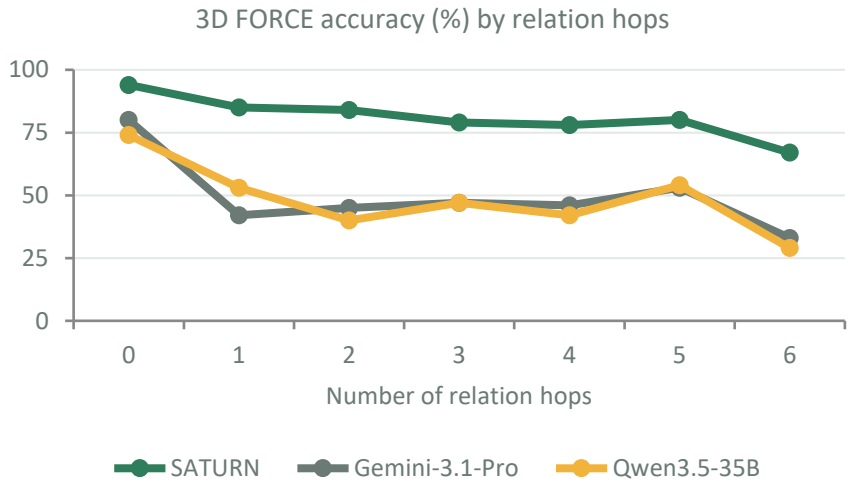
```
is_notebook(x)
& obj_left(x, robot)
& obj_front(x, me)
```

✓ n2

**Paper** *Kamali, Premsri, **Rajpal**, Zadeh, Li & Kordjamshidi. SATURN: Symbolic Spatial Reasoning for Multi-Perspective Grounding. EMNLP 2026.*

# With symbolic representations of images

### STABLE ACROSS INCREASING COMPLEXITY




### MORE ACCURATE

**+25.1%** vs. 57.8% for the best VLM on 3D FORCE

**→ Training-free, more accurate and more stable than end-to-end VLMs**

**Paper** *Kamali, Premsri, **Rajpal**, Zadeh, Li & Kordjamshidi. SATURN: Symbolic Spatial Reasoning for Multi-Perspective Grounding. EMNLP 2026.*

# Beyond spatial reasoning



## Spatial

Stories, scenes

↓

**Grids, 3D scenes, programs**



## Legal

Contract clauses

↓

**Rule graphs with conditions & exceptions**



## Temporal

Event descriptions

↓

**Event logic trees over time**



## Structured data

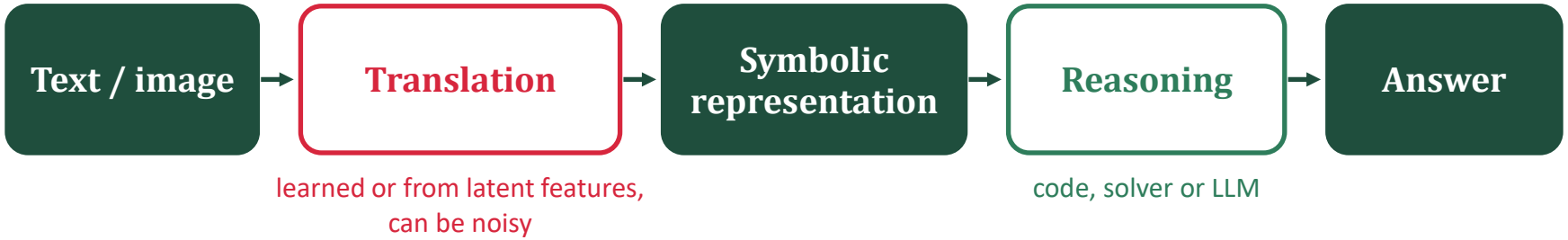
Questions

↓

**SQL, knowledge graphs**

**Same recipe:** translate → structure → execute. **Same risk:** a wrong structure still executes.

# Open question: the translation step




Our grids: 100 F1 from gold relations, 87–90 F1 from extracted relations (StepGame)

**How do we improve the translation?**

**How do we build metrics to test it without gold structures?**

Lightning talk 1 of 3



# Transforming Interpretability Gains to Explainability Wins


## Gokul S Krishnan

Senior Research Scientist, Centre for Responsible AI, IIT Madras


*5 min talk + 5 min discussion*

Lightning talk 2 of 3

# LEGO: Synergizing Expert GraphRAG and Expert Chain-of-Thought for Legal Reasoning

**Qingjing Chen**
PhD candidate, University of Bologna
EMNLP Findings 2026

*5 min talk + 5 min discussion*

Lightning talk 3 of 3

# Grammar of the Wave: Towards Explainable Multivariate Time Series Event Detection via Neuro-Symbolic VLM Agents

**Sky Chenwei Wan**
PhD Student, SLB, France
EMNLP 2026

*5 min talk + 5 min discussion*

# Panel discussion

1. How can neuro-symbolic AI make reasoning more trustworthy?
2. When do rules and programs help, and how should they pair with foundation models?
3. Beyond accuracy, what evidence shows that reasoning is trustworthy?
4. Where should neuro-symbolic and trustworthy reasoning go next?

Daniel Preotiuc-Pietro · Leslie Barrett · Parisa Kordjamshidi · Artem Shelmanov

# Thank you

Questions and open discussion

**Panel:** Daniel Preotiuc-Pietro · Leslie Barrett · Parisa Kordjamshidi · Artem Shelmanov

**Lightning talks:** Qingjing Chen · Gokul S Krishnan · Sky Chenwei Wan

**Organizer:** Shreya Rajpal, Michigan State University

